

Layer Structure Guide — 14-Layer Toggleable Hierarchy

Every v2 SVG uses a strict bottom-to-top stack of **14 Inkscape layers** (`inkscape:groupmode="layer"`), so each can be toggled independently in Inkscape, Illustrator, or any CAD tool that honours SVG layer groups.

Layers are emitted in draw order: **Layer 0 is drawn first (bottom), Layer 13 last (top).**

#	Layer label	Contents	Notes
0	Background	Sheet fill, outer/inner frame, title block, header strip, version stamp	Always on
1	Scope	Scope-boundary diamonds (TPXXXXY), terminal/handover points	Defines the “last-meter” interface
2	Structures	Cavity geometry, outer shell, vacuum vessel	Rendered light grey as background context
3	Equipment	Vessels, heat exchangers, cavities, couplers, tuners, tanks	Major equipment glyphs
4	Piping-40 K	40 K thermal-shield circuit	Red (#e00000)
5	Piping-4.5 K	4.5 K supply circuit	Blue (#0033cc)
6	Piping-2 K	2 K return circuit	Cyan (#00a6bd)
7	Piping-Water	DI cooling-water / utility circuit	Green (#00a000)
8	Valves	All valve symbols + 60 K guard / infrastructure / instrument-air services	Manual, control, safety, relief, MV bellows
9	Instruments	ISA sensor bubbles (TT, PT, LT, FT, ...)	Field-mounted discrete instruments
10	Control	DIS interlock block, tuner limit switches, Lemo connectors, actuators	Control / interlock logic
11	Signals	Signal lines + heat-load callouts	Pneumatic / electric signal layer
12	Tags	All instrument tags, valve tags, buffer &	Text layer

#	Layer label	Contents	Notes
		hand-over annotations	
13	Legend	Drawing legend (classes, symbols, scope keys)	Always on

Emphasis behaviour by sheet type

- **Process / Cryogenic sheets (Sheet 1):** piping layers 4-7 are drawn at full weight & colour; only inline process devices (FT/PT/LT) appear on the instrument layer. Control layer (DIS etc.) is empty.
- **Instrumentation sheets (Sheet 2):** piping layers 4-8 are drawn as a faded grey backdrop so the full sensor suite, control layer, and signals stand out.

Style-version interaction

The two style versions modify rendering weights/colours within these layers (see Drawing Index §2) but do **not** change the layer structure — the 14-layer hierarchy is identical across both `_STANDARD` and `_CONTROL-CENTRIC` files.